

THIRD
STEP

Third Step

Fish Trading

E-Commerce Platform · Admin Backend · Firebase

Project Type:	Full-Stack Web Application
Platform:	Firebase Hosting · Firestore · Auth
Deployment:	freshcatch-uae.web.app
Date:	June 2026
Version:	1.0.0

OVERVIEW

Executive Summary

Third Step Fish Trading is a fully responsive, Arabic-first e-commerce web platform purpose-built for fresh seafood retail in the UAE. The platform enables customers to browse live fish inventory, select custom cuts, add delivery preferences, and place orders directly via WhatsApp — all within a single-page application. An integrated Firebase-powered admin panel allows the business to manage products, process orders, upload daily catch, manage categories, and print kitchen tickets.

4	5	3
Pages / Sections	Admin Modules	Firebase Services
Shop · Catch · How It Works · Our Story	Dashboard · Products · Orders · Catch · Categories	Hosting · Firestore · Authentication

TECHNOLOGY

Technology Stack

Frontend	Vanilla HTML5, CSS3 (custom properties), JavaScript ES Modules
Fonts	Tajawal (Arabic / RTL), Playfair Display (serif headings), Inter (body)
Backend / DB	Firebase Firestore (NoSQL, real-time) — collections: products, orders, categories, settings
Authentication	Firebase Auth — Email / Password for admin panel login
Hosting	Firebase Hosting — CDN-delivered at freshcatch-uae.web.app
Maps / Location	OpenStreetMap embed + Nominatim reverse geocoding (no API key required)
Messaging	WhatsApp deep-link integration — structured order messages sent via wa.me
Responsive	Mobile-first CSS — breakpoints at 768 px and 390 px
RTL Support	Full dir="rtl" support with Tajawal font; toggle stored in localStorage
Print	CSS @media print — kitchen ticket receipt layout for POS printing

FRONTEND

Customer-Facing Website Features

3.1 Navigation & Global UI

- Sticky top navigation with logo, page links, search, language toggle, and cart button
- Cart button collapses to icon-only on mobile to save space
- Full-screen mobile hamburger menu with category shortcuts and delivery info
- Arabic / English RTL■LTR toggle persisted across sessions via localStorage
- Live search with instant dropdown results filtered by fish name or Arabic name

3.2 Shop Page

- Hero section with animated fish preview cards (Hamour, Kingfish, Shaari) and horizontal scroll-snap
- Sidebar category filter (desktop) — All, Large fish, Small fish, Shellfish, Shrimp, Pre-orders
- Horizontal chip bar (mobile) for category filtering
- Sort toolbar — Popular, Price low–high, Price high–low, Name A–Z, Pre-orders
- Product cards with emoji fish image, Arabic name, English name, family name, price/kg, and stock badge
- Real-time item count badges per category in sidebar
- "Best seller", "Pre-order", "In stock", "Low stock" dynamic badge system

3.3 Product Detail Page (PDP Modal)

- Slides up as bottom sheet on mobile; side panel on desktop
- Gross weight and net yield tags (e.g. "1.4 kg gross → 1.0 kg net")
- Dynamic stock CTA — "Add to cart" / "Pre-order" / "Out of stock" based on stockQty
- 3×2 cut grid with emoji icons — Whole, Head-off, Fillet, Steaks, Cubes, Cleaned
- Add-on fast taps — Ice pack, Gift box, Sayadieh rice, Spice mix
- Free-text notes field forwarded to kitchen

3.4 Cart & Checkout

- Slide-in cart drawer with quantity controls, per-item cut and add-on summary
- Cart subtotal includes add-on prices; delivery fee added at checkout
- Delivery type selector — Home Delivery (AED 15) or Shop Pickup (Free)
- Customer form — Full name, Mobile (WhatsApp), Delivery address
- Geolocation detection + OpenStreetMap embed for address confirmation
- Delivery window calendar — 3-day grid with AM/PM slots; pre-order dates blocked
- Driver notes field
- Live WhatsApp message preview updates as form is filled
- Structured order ID generated: #FT-YYYY-XXXX
- One-tap "Place order via WhatsApp" button sends full formatted order

3.5 WhatsApp Order Message Format

Each order is formatted as a structured WhatsApp message containing: Order ID, delivery type, itemised fish list (with cut, notes, and add-ons), customer details, map pin link, chosen delivery window, and a weight disclaimer. Pre-order items are flagged with [PRE-ORDER].

3.6 Other Pages

- Catch of the Day — featured fish highlights, availability grid, freshness promise strip
- How It Works — 4-step illustrated guide, cuts reference grid, WhatsApp preview demo
- Our Story — brand narrative, sustainability stats, team grid, contact map
- Footer — Third Step Fish Trading branding, navigation links, contact info

ADMIN PANEL

Admin Backend — Feature Modules

4.1 Authentication

- Firebase Email / Password login with error handling
- "Continue in demo mode" bypass for testing without credentials
- Demo mode notice banner shown when Firebase credentials are placeholders
- Auto sign-out on session expiry; user email displayed in sidebar footer

4.2 Dashboard

- Live stat cards — Total Products, Total Orders, Pending Orders, Total Revenue
- Recent orders table with status badges and quick-view links
- Date/time header updated in real time

4.3 Products Management

Add Product	Modal form: name, Arabic name, category, price, unit, stock qty, badge, gross/net weight, cuts, image URL, pre-order flag
Edit Product	Pre-filled modal; all fields editable; saves to Firestore or updates DEMO_PRODUCTS in demo mode
Delete Product	Confirm dialog before removal to prevent accidental deletion
Search & Filter	Real-time search by name; filter by category and stock status
Category Sync	Product category dropdown rebuilds dynamically when categories are added/edited

4.4 Orders Management

Order List	Full table — Order ID, customer name, items, total, status, date
Order Detail	Click-to-expand modal with full order breakdown including cuts, add-ons, notes, address, delivery window
Status Update	Change order status: Pending → Confirmed → Preparing → Out for Delivery → Delivered
Actual Weight	Weight adjuster bar — admin enters scale weight in kg; revised total auto-calculated and saved to Firestore
Kitchen Ticket	One-click print — generates receipt-style ticket with fish name, cut, and customer notes in UPPERCASE bold

Delete Order	Confirm before delete; updates order count in dashboard stats

4.5 Catch Upload (POS)

- Form to upload daily fresh catch — fish name, Arabic name, price/kg, gross weight, net yield, stock qty, category, badge
- "Push Live" button publishes catch item to Firestore with timestamp and elapsed-time counter
- Recent Catches list shows items uploaded in current session with live elapsed timer
- Designed for quick morning dock-to-site workflow

4.6 Categories Management

Add Category	Name, Arabic name, slug (URL-safe identifier), and colour picker
Edit Category	All fields editable; slug locked after creation to prevent breaking product links
Delete Category	Confirm before delete; updates product dropdowns and mobile menu automatically
Live Sync	All product category dropdowns and the customer-facing mobile menu rebuild on every add/edit/delete

4.7 Settings

- Store info — business name, WhatsApp number, address
- Delivery settings — fee, available windows, cut-off time
- Holiday / closure toggle
- Changes saved to Firestore settings collection

Mobile Responsiveness & RTL Support

5.1 Mobile Breakpoints

≤ 1024 px (Tablet)	Narrower sidebar, reduced hero padding, 2-column cuts grid
≤ 768 px (Mobile)	Sidebar hidden → chip bar; cart icon-only; full-screen modals slide up from bottom; 2-column product grid; checkout stacks vertically
≤ 390 px (Small)	Tighter padding, 24 px hero heading, single-column values grid

5.2 Admin Mobile Layout

- Sidebar collapses into a fixed bottom tab bar (icon + label) — mimics native app navigation
- Main content gains bottom padding to clear the tab bar
- Stat cards switch to 2-column grid
- Data tables get horizontal scroll with momentum scrolling (-webkit-overflow-scrolling: touch)
- Modals slide up as bottom sheets; form buttons stack full-width

5.3 RTL (Arabic) Support

- dir="rtl" applied dynamically via JavaScript; no page reload required
- Preference stored in localStorage — persists across browser sessions
- Tajawal Arabic font loaded for all body text in RTL mode
- All flex layouts, border sides, absolute positions, and margin/padding directions flip correctly
- Checkout form inputs switch to direction:rtl; WhatsApp preview renders right-to-left
- Admin sidebar toggle button (🏠 / EN) visible in sidebar footer
- Bottom tab bar remains functional in RTL mode on mobile

FIREBASE

Firestore Architecture

6.1 Firestore Collections

products	id, name, nameAr, family, category, price, unit, stockQty, grossWeight, netYield, badge, imageUrl, preOrder, cuts, createdAt
orders	id, orderId (#FT-YYYY-XXXX), customer, mobile, address, geoPin, items, addons, notes, deliveryType, deliveryWindow, subtotal, delivery, total, actualWeight, revisedTotal, status, createdAt
categories	id, name, nameAr, slug, color, createdAt
settings	storeInfo (name, whatsapp, address), delivery (fee, windows, cutoff), holiday

6.2 Firebase Services Used

- Firestore — NoSQL document database for all product, order, category, and settings data
- Firebase Auth — Email/Password sign-in for admin panel; only authenticated users can write
- Firebase Hosting — Static site CDN with custom rewrites (/admin → /admin/index.html)
- Firebase SDK v10.12.0 loaded via CDN ES modules (no build tool required)

6.3 Fallback / Demo Mode

If Firebase credentials are not configured or authentication fails, the app gracefully falls back to local DEMO_PRODUCTS and DEMO_ORDERS arrays. The admin panel shows a "Demo mode" banner and all CRUD operations update in-memory state. This allows full UI testing without a live Firebase project.

ORDER FLOW

End-to-End Order Flow

	Customer browses	Shop page loads products from Firestore (or demo data). Filter by category, sort, or search by name.
2	Product detail	Click card → PDP modal: view gross/net weight, select cut, choose add-ons, add notes.
	Add to cart	Item added to CART array with cut, notes, and addons. Cart badge updates.
4	Checkout	Fill name, mobile, address (or detect location), select delivery type and window.
	WhatsApp order	Order ID generated (#FT-YYYY-XXXX). Structured message built and sent via WhatsApp deep link.

	Admin receives	Order appears in admin Orders panel with status "Pending".
7	Kitchen ticket	Admin prints kitchen ticket — fish name, cut, notes in uppercase bold for kitchen staff.
	Weight adjust	After weighing fish, admin enters actual kg. Revised total calculated and saved.
9	Status update	Admin updates status through: Confirmed → Preparing → Out for Delivery → Delivered.

DEPLOYMENT

Deployment & Configuration

8.1 Live URLs

Customer Site	https://freshcatch-uae.web.app
Admin Panel	https://freshcatch-uae.web.app/admin
Firebase Console	https://console.firebase.google.com/project/freshcatch-uae

8.2 File Structure

- index.html — Customer-facing SPA
- css/style.css — All frontend styles including mobile breakpoints and RTL overrides
- js/app.js — Firebase init, product rendering, cart, PDP modal, checkout, WhatsApp message builder
- admin/index.html — Admin panel shell with all module pages
- admin/css/admin.css — Admin styles with mobile bottom tab bar and RTL overrides
- admin/js/admin.js — Firebase Auth, Firestore CRUD, categories, catch upload, kitchen ticket printer
- img/logo.png — Third Step Fish Trading brand logo
- firebase.json — Hosting config with /admin rewrite rule
- .firebaserc — Project binding to freshcatch-uae

8.3 Setup Checklist

1	Firebase project created (freshcatch-uae)	✓ Done
2	Firebase credentials added to js/app.js and admin/js/admin.js	✓ Done
3	Firebase Hosting deployed	✓ Done
4	Enable Firebase Auth — Email/Password sign-in method	Pending
5	Create admin user in Firebase Auth console	Pending
6	Create Firestore database (test mode → lock down rules)	Pending
7	Update WhatsApp number in js/app.js (WA_NUMBER variable)	Pending